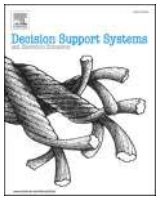
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Delegation of purchasing tasks to AI: The role of perceived choice and decision autonomy

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ABSTRACT

Although artificial intelligence (AI) outperforms humans in many tasks, research suggests some consumers are still averse to having AI perform tasks on their behalf. Informed by the literature of customer decision-making process, we propose and show that consumer autonomy is a significant predictor of customers' decision to adopt AI in the purchasing context. Across three experiments, we found that the delegation of purchasing tasks to AI, which restricts *choice* and *decision* dimensions of consumers' perceived autonomy, reduces the likelihood of AI adoption. Our results show that the effects of choice and decision autonomy on AI adoption holds even when product choice evaluation is complex. We also found that identity-relevant consumption moderates this relationship, such that it interacts with choice and decision autonomy. Specifically, despite lacking choice and decision autonomy, those who identify strongly with a given activity are more likely to use an AI-enabled app to purchase the product needed to perform this activity. With these insights into when and why consumers are likely to use AI-enabled technology, firms might effectively increase its adoption.

1. Introduction

Artificial intelligence (AI) refers to "programs, algorithms, systems and machines" [1,p. vi] that "mimic human intelligence in tasks such as learning, planning, and problem-solving through higher-level, autonomous knowledge creation" [2,p. 93], using technologies such as machine learning and deep learning, automation, neural networks, and rule-based expert systems [3]. Currently, AI has been widely applied to facilitate customer purchase decision-making process through recommender systems[1] (RS) and autonomous shopping systems[2] (ASS). In essence, AI-enabled RS and ASS are decision support systems (DSS) that perform some or substantial parts of decision-making for users as to which products and services to select, purchase, download, or consume [4,8]. In this context, the use of AI is intended to reduce consumers' effort and uncertainties [9,10] by enabling customers to find items that match their preferences more efficiently [11,12] and achieve higher quality purchase decisions [13,14]. However, customers may perceive such aids as threats to their decision-making autonomy, resulting in resistance to adopt them [4,15,16].

In RSs and ASSs, AI takes over two customer roles. First, by generating recommendations, it narrows choices to consideration sets on customers' behalf. More precisely, the information collection and choice reduction stage of the decision-making process traditionally performed by customers are delegated to AI, such that it filters options and prioritizes the order of choices exposed to customers [17]. Thus, AI's algorithm shapes customers' choice sets, which eventually determines their preferences and decisions [18]. For example, Facebook and Instagram show content that is predicted by AI to be the most engaging, rather than making all subscribed posts visible on a user's newsfeed. Consequently, users are unable to engage with posts that AI has excluded from their consideration sets. The "freedom to do otherwise," also referred to as free will, is not available to such users [19]. Thus, customers' perceived *choice autonomy,* which we define as their subjective sense of being able to view and narrow down options freely without external influence, is

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[1] Recommender systems provide personalized recommendations according to algorithmic performance [4,5]. These recommendations can be based on a variety of algorithms, such as user-based collaborative filtering [6], previous purchases and users' data availability, which can be collected explicitly or implicitly [7].

[2] Autonomous shopping systems are defined as "technology to which consumers can delegate substantial parts of the shopping process, including shopping decisions and tasks. The system autonomously reaches a series of conclusions for consumers, such as which and how many items to buy and when to do so, based on input data" [4,p. 75].

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reduced significantly [15,16]. Furthermore, in many instances, consumers are not fully informed about the parameters and corresponding weights that underlie algorithms' recommendations [20]. In addition, due to their ability to improve without being explicitly programmed, some machine learning algorithms that are used to increase prediction accuracy for customers' preferences are inscrutable even to the engineers who initially developed them [21]. Thus, we argue that AI's function of narrowing choices to consideration sets significantly affects consumer perceived choice autonomy.

Second, AI has the ability to go beyond filtering choices and actively take over final purchasing decisions, without consumer intervention. This action diminishes customers' perceived *decision autonomy*, that is, their subjective sense of being able to make up their own minds about whether to buy products and which brands or products to purchase [16,22]. For example, smart refrigerators can feed data to an algorithm that determines when to restock and what to purchase. Companies like Boxed's Concierge already practice this type of shipping-then-shopping business model, in which the customer's final decision gets relinquished to AI. Similarly, Spotify, YouTube and Netflix use AI to determine the content that will be auto-played.

Despite the prevalence of AI in the consumer decision-making process and scholars' warning that its use may reduce customers' sense of autonomy [4,15,16], empirical evidence remains scarce. Exceptions include findings indicating that receiving help from AI reduces computer game enjoyment because it undermines players' autonomy [23], and that individuals prefer imperfect algorithms they can modify to assert control over forecasting outcomes [24].

While perceived autonomy is not a new concept in the RS literature, the discussion largely revolves around users' ability to control their interaction with the systems. Examples include if the RS allows users to offer inputs such as rating, attribute preference, revise preferences or skip input responses [25,26]. Having control over interactions with RS increases positive attitude [25], reduces perceived risks [27], and enhances user experience [28]. Other user controls of RS (e.g., allowing users to request for "show more popular items") result in more positive evaluation of the recommendations [29]. When users are allowed to select the type of algorithm, they are likely to exercise that option and use the RS in the long-term [30].

User autonomy in the RS literature is also explored through the methods in which customer preferences are elicited (i.e., explicit preference elicitations imply greater user autonomy) and how they affect decision quality [31]. RS users are more likely to accept top-ranked recommendations (resulting in better decision quality) when their preferences have been elicited explicitly (i.e., rate the desirability of product attributes and their importance) as opposed to a less transparent method [32]. Equally, query-based RS (i.e., user assigns weights to each of the attributes) increases the likelihood of user preference matching customers' choice and boosts their decision confidence [33]. Prior studies also examine the effectiveness of RSs asking users about their product-related needs rather than their specifications of product attributes [34]. In sum, prior studies mainly focus on the types or the extent of control users have when providing their preference input and how they affect satisfaction and the decision outcomes.

However, as RSs become more autonomous, the interaction between RSs and users, as well as how consumer preference is gathered, have evolved from the context of prior literature. Currently, consumers typically have little control over the actual inputs considered by the RS (e.g., consumer preferences), its processes (e.g., algorithm for generating recommendations), or output (e.g., list of recommendations offered [35]). Thus, scholars have called for a better understanding on the consequences of awarding more autonomy to an actor (i.e., AI in RS and ASS) on another actor (i.e., consumers [36]). Specifically, we need to update our knowledge as to whether the relinquishment of customers' perceived autonomy in their decision-making has a consequential impact on their use of the technology.

To address the knowledge gaps, we examine whether reducing customers' perceived autonomy of *choice* and *decision* through delegation of some or all aspects of purchasing tasks to AI affect AI adoption. Our research seeks to answer two key questions. First, *in a consumer purchase decision-making setting, how do perceived choice and decision autonomy affect the likelihood that consumers adopt AI*? Second, *what are the boundary conditions of these effects*?

Thus, we make several theoretical contributions. First, our empirical examination of perceived autonomy as a determinant of AI adoption contributes to research on human aversion to AI. Prior studies indicate that resistance to AI stems from factors such as lack of trust [4], characteristics and consequences of the tasks [37,38], customer characteristics [38], and the salience of the task to customer identity [39]. Furthermore, individuals have preconceived expectations when interacting with AI [40] and hold the perception that AI is less able to cater to customers' unique features and is less competent in the hedonic consumption domain [41].

Second, we update the discussion of user autonomy in the decision support systems field beyond the scope of user ability to control their interaction with RSs and how they express their customer preferences. As DSSs are designed to become more autonomous and consumers have less control over the actual inputs used by the systems (e.g. consumer preference, [35]), our research sheds light on whether delegating the decision-making autonomy in purchasing is desirable in exchange for the convenience they provide. In addition, our investigation of autonomy also enriches the decision support systems literature that, to date, identifies factors such as perceived usefulness, attitudes, trust, and preferences as determining factors of DSS tools adoption (e.g., [7]). Third, by breaking down perceived autonomy according to task delegations to AI along the consumer decision-making process (i.e., narrowing consideration sets and making final decisions), we explore the levels and types of autonomy that customers find desirable to retain (vs. relinquish) at different stages of the purchase journey [42]. Fourth, by examining the boundary condition of the perceived autonomy–AI adoption relationship in which the negative effects of lack of autonomy are attenuated, we inform optimal designs for hybrid human–AI decision-making processes that can balance its benefits and risks [15].

2. Conceptual background and hypotheses development

2.1. Perceived autonomy

Perceived autonomy is an "individual's subjective sense of being able to make and enact decisions of their own volition" [16,p. 431]. In the consumer choice context, it refers to customers' abilities to develop purchase considerations that are not imposed on them by external parties but are authentic to themselves [15,16]. External forces such as coercion, manipulation, available options, laws, regulations, and social pressure can influence customers' purchase decisions. For example, in online shopping environments, the display order of attributes in product recommendations also influences those attributes' importance, which affects final choices [43]. Scholars have shown that customers are vulnerable to biased product recommendations from RSs that are perceived as personalized, which results in sub-optimal product choices [35].

Traditionally, the consumer decision-making process involves various stages, such as gathering information about alternatives, comparing those alternatives, and deciding which alternatives to purchase [44]. Even during pre-AI era, consumers had been delegating some of their purchase decisions to agents (e.g., bridal consultants, decorators). Customers may choose to delegate certain or all parts of the decision-making process (e.g., information gathering, choice set reduction, final choice decision), depending on the extent to which they perceive they retain or lose their autonomy [45]. Because AI may be engaged in different stages of the decision-making process, it implies different outcomes for consumer autonomy.

2.2. Choice autonomy

Within the boundaries of customers' resources,[3] consumers tend to use a two-stage approach when faced with extensive options [44]. In the first stage, they screen a set of available products to identify subsets of options that are worth including in their consideration sets [46]. In the second stage, to make their final choices, they evaluate the screened subsets in more detail, by performing relative comparisons of important product attributes across available alternatives [13].

In recent years, the task of narrowing consideration sets (the first stage of decision-making) often has been delegated to AI-enabled RSs that offer personalized recommendations based on certain parameters. RSs are prevalent in marketplaces that involve large sets of alternatives, such as online shopping (e.g., Amazon), movies (e.g., Netflix), music (e.g., Spotify), financial investments (e.g., Wealthfront), and dating (e.g., Tinder). The parameters applied may include customers' stated preferences, past behaviors, comparisons with peer users, and current trends [47]. However, they are not always based on consumers' chosen preferences, nor are they completely known to customers [20,21]. Thus, not only do customers lack the autonomy to choose to view all available options, they also lack the power to decide on the screening parameters. As the options in the consideration set are pre-determined by AI, the final choices made in the second stage are largely limited to those initial choices presented by AI. Perceived choice autonomy is customers' subjective sense of being able to view and narrow down options freely without external influence [15,16]. Accordingly, when AI narrows down options based on parameters unknown to customers, it takes over the autonomy that customers exercise in the first stage of the decision-making process, or their choice autonomy.

When choice screening is done by AI, depriving them of their choice autonomy, customers may feel that they end up with suboptimal options due to the algorithm's incorrect assumptions and mismeasurements [48]. For example, RSs might wrongly assume customers always want to consume similar items (e.g., same genre of books, films, or music) when in reality, occasionally they prefer different options [20]. Furthermore, the algorithm might not include parameters that customers regard as important, such as sustainability. Customers' sustainability, cultural, social, and ethical concerns have not yet been considered in the literature as the main inputs to build customers' personalized models for RSs [49]. Suboptimal recommendations also can arise if the algorithm recommends an item purchased by similar customers despite the item being an inferior choice (e.g., overpriced laptop) [50]. Furthermore, noise in the rating data negatively affects the performance of collaborative filtering, which then reduces the predictive accuracy of RSs [6].

Customers may also perceive that they are deprived of the opportunity to develop or change their preferences over time [15]. AI generates recommendations according to past behaviors, which leads customers' choices to become more predictable, self-perpetuating, and path-dependent than their own choices would have been [16]. Finally, as the recommendations proposed by an algorithm may include parameters that serve firms' strategic interests, rather than customer welfare, customers may perceive that their choice autonomy is being restricted by firms. For example, Netflix's algorithm manifests its strategy to promote more exclusive content rather than classic movies. In this sense, Netflix influences customers' preferences through the content it pushes [51] rather than matching recommendations with customers' preferences.

[3] Even if customers have the freedom to view all options and choose from those options, it may not be possible for them to view and access all the choices available, because they may be constrained by their own resources (e.g., time, cognitive capacity) or information availability. These constraints arise for everyone, to varying degrees.

2.3. Decision autonomy

Whereas choice autonomy relates to customers' abilities to view available choices and narrow them into consideration sets without external influences, such as algorithms, decision autonomy refers to customers' power to make the final choices and execute their purchases without being externally influenced. It reflects the second stage of the decision-making process [44]. When customers delegate the task of making the final decision to agents that conduct the purchases on their behalf, such as in ASSs, they relinquish the decision dimension of autonomy. ASSs that rely on AI algorithms can go beyond recommendations and take over the shopping process [4]. They determine autonomously which and how many items to buy and when, according to predictions developed by algorithms. For example, Amazon holds a patent for anticipatory shipping, a method to deliver products even before customers make the purchases [52]. Similarly, Boxed's Concierge service ships products without customers' intervention, according to AI predictions of when customers will run out of the items [53]. Box subscription companies such as Trendy Butler have AI decide which items of clothing to be included in the box for customers' subscriptions. Furthermore, firms are developing AI-enabled smart fridges and laundry machines that have automated purchasing capabilities or can be linked to other smart apps such as recipe planning and grocery shopping [54,55]). In the digital context, the algorithm of Netflix takes over customer decision autonomy by auto playing the next episode of a show. Meanwhile, YouTube and Spotify auto-queue content based on real-time consumption.

While customers retain their rights to return unwanted products, cancel their subscriptions, or stop the auto-play of YouTube and Spotify, this power lies outside the decision-making purchase process. Consumer decision-making process starts from the stage of problem recognition (the moment a customer realizes that there is a need for a product), leading to information search, evaluation of alternatives, making the final selection, and post-purchase evaluation (e.g., [56]. Product returns and service cancellations occur post decision-making as a result of customers' evaluation of their satisfaction with the product [56] or the decision-making process that they experienced [57]. For example, once YouTube auto-plays content that the algorithm dictates, we argue that the decision has been made for consumers. The freedom to continue or stop consuming the content is dependent on their satisfaction with the AI-chosen content and/or their satisfaction with how the decision process was performed for them by AI.

It is important to note that surrendering decision autonomy to external agents does not mean relinquishing choice autonomy; customers may perform the first stage of the decision-making process by searching for information, comparing alternatives, and narrowing lists of acceptable alternatives before delegating the task of making final decisions [45]. They might view and compare all possible alternatives, put the items in their wish lists (their consideration sets), and pass the lists on to AI to make the final choices. As AI agents learn from consumers' purchasing patterns, they also may take over decision autonomy, by executing automated repeat purchases of products for which customers performed the initial task of narrowing choice.

2.4. Hypotheses development

2.4.1. Choice and decision autonomy on AI adoption

A key function of AI is to narrow down large numbers of available options to relevant consideration sets. A large number of options may increase decision difficulty and preference uncertainty [58], but it also increases freedom of choice [59], which brings several benefits to customers. Greater assortments increase the likelihood that consumers can find ideal options that match their preferences [60] and fulfill their variety-seeking needs (e.g., 61]. They also offer greater flexibility when customers experience uncertain preferences for future choices [62]. Finally, choice enhances intrinsic motivation, perceived control, and

evaluations of chosen alternatives, which improves consumer well-being [63]. Thus, restrictions on customer choice autonomy may have negative effects on immediate and future decision outcomes.

The act of choosing and deciding provides psychological value to a person; for example, exercising their decision autonomy enables customers to reflect on and reform their preferences [20]. By exercising their autonomy to choose, they can learn from their past decisions and change them according to their long-term goals [64]. Moreover, customers' abilities to decide shape and reflect their identities, personal values, and views (e.g., books and newspapers one chooses to read [65]), and create and boost their self-image. Possessions and purchases play important roles in creating one's sense of identity (e.g., [66]). When customers choose products that differ from others' preferences, they make strategic moves that signal their identities [67].

The ability to exercise decision autonomy is also important because it allows people to take ownership and attribute favorable outcomes to their own actions [68]. Making one's own decisions reduces the risk of dissonance by boosting evaluations of chosen options [69]. People derive positive affect from personally causing certain outcomes. For example, Hui and Bateson [70] show that in service encounters, decision autonomy enhances consumers' perceived control, which creates positive affect and behavioral responses. This finding likely explains why consumers report less enjoyment of a computer game when help is provided [23].

Exercising one's autonomy to choose generates positive affect and heightens motivation [71], whereas restricting it undermines motivation and evokes psychological reactance (i.e., a motivational state focused on reattaining the restricted freedom [72]). For example, perceived loss of autonomy inhibits purchasing decision delegation to human agents [45]. When consumers' preferred options are out of stock at the stores, they feel less satisfied with their decision processes and are less likely to return to those stores [73]. Moreover, upon learning their choices are predictable, customers assert their autonomy by choosing less-preferred options [74]. Because autonomy prompts pleasurable feelings that lead to well-being, we predict that perceptions of choice and decision autonomy retained by customers will function as antecedents of AI adoption.

H1. The lower the (a) perceived choice autonomy, and (b) the perceived decision autonomy, the lower the AI adoption likelihood.

2.4.2. Identity as a boundary condition of autonomy – AI adoption relationship

Consumption decisions are often derived from consumers' desire to conform to their identity [75]. As identity salience increases, it is more likely that the identity has an effect on a consumer's attitude and behavior [75]. Prior studies have highlighted that technologies can undermine people's identities when they take over functions that are pertinent to their self-concepts, resulting in technology aversion. Performing identity-relevant tasks such as fishing, driving, or baking requires specific skill sets, which allow internal attribution (i.e., people crediting their own actions as the cause of outcomes [39]. Customer resistance occurs when the delegation of processes to technologies hinders the attribution of identity-relevant consumption outcomes to themselves [39]. In contrast, technologies such as AI do not always prevent consumers from performing functions that are central to their identities; they can also enable consumers to reinforce those identities.

A product will be considered by consumers if they are goal-relevant; that is when they are linked to an outcome that is important to the individual's identity [75]. For example, avid runners with specific monthly mileage goals are more likely to use running apps that track their runs. Similarly, a product will be deliberated positively if it allows an individual to perform behavioral functions associated with a particular identity; i.e., a baseball player requires a bat, glove, and cleats to fulfill that identity [76]. As such, an AI-enabled purchasing app that efficiently purchases the products required for the person to perform that identity is considered goal-relevant because it helps to achieve the outcome that is important to their identity. As AI takes over parts or all of the purchasing process of products needed to perform an identity-salient activity (e.g., running shoes for avid runners), it complements the runner in achieving his goal and in so doing, allows them to devote their time and energy to the actual activity (e.g., running). We predict that in this case, the negative effect of the lack of choice and decision autonomy on the likelihood of AI use is likely to be attenuated. Thus, we hypothesize the following:

H2. Identity salience to the activity interacts with (a) choice autonomy, and (b) decision autonomy such that higher (vs. lower) identity salience attenuates the negative impact of the lack of autonomy on the AI adoption likelihood.

3. Method

3.1. Study 1: effects of choice and decision on adoption

In Study 1, we use an experimental approach to test H1: the individual impact of the two dimensions of autonomy—choice and decision—on the adoption of AI-enabled technology.

3.1.1. Design, participants, and procedure

In the first study, participants were randomly assigned to one of four conditions. We use a 2 (choice autonomy: present vs. absent) x 2 (decision autonomy: present vs. absent) between-subjects experimental design. "Present" indicates that the participant retains the autonomy while "absent" signifies that participant relinquishes the autonomy to AI. The manipulations involve a grocery shopping app that enables consumers to shop on their mobile phones. We informed participants that the goal of the app was to make their purchase process easier.

The scenarios describe whether participants can see all grocery choices available; i.e., choice autonomy present (vs. not able to see all the available choices; i.e., choice autonomy absent) and whether they make the final choices and execute the purchases themselves; i.e., decision autonomy present (vs. the app makes the final choices and executes the purchases; i.e., decision autonomy absent). As the dependent variable, we used adoption of the AI technology, that is, the likelihood of the participant using the app. Appendix A presents details of the manipulations, measures used, and participant frequency per condition. A total of 535 Prolific-registered panel members residing in the United States (M_{Age} = 32.78 years; 64.7% women) completed the study for monetary compensation.

3.1.2. Results and discussion

3.1.2.1. Manipulation checks. We adapted prior literature to develop our manipulation checks [77]. Specifically, we asked participants to answer two questions: "How much control do you feel over the grocery choices you get to see?" and "How much control do you feel you have over the purchase decision of grocery products?", both using a 7-point Likert scale (1 = "do not have control at all," 7 = "have a lot of control"). Participants assigned to the choice autonomy-present condition indicate greater control over their choices than those in the choice autonomy-absent condition (M_{Choice} = 4.89; SD = 1.76; $M_{NoChoice}$ = 2.74; SD = 1.4; $t(533) = -15.68$, $p < .001$; *Cohen's d* = 1.35). Those assigned to the decision autonomy-present condition indicate greater control over their purchase decisions than those in the decision autonomy-absent condition ($M_{Decision}$ = 5.23; SD = 1.58; $M_{NoDecision}$ = 3.11; SD = 1.73; $t(533) = -14.78$, $p < .001$; *Cohen's d* = 1.28).

3.1.2.2. Adoption of AI. An ANOVA analysis features choice autonomy and decision autonomy as independent variables and AI adoption as the dependent variable. The main effects of choice autonomy ($F(1, 531) = 27.83$, $p < .001$, $\eta^2 = 0.05$) and decision autonomy ($F(1, 531) = 67.22$, p

$< .001$; $\eta^2 = 0.112$) on AI adoption are significant. These results provide support for H1. While H1 focuses on the individual effects of autonomy on AI adoption, we also tested for the interaction effect for a robustness check. We find that choice and decision autonomy interact to affect customer AI adoption ($F(1, 531) = 4.13$, $p = .043$; $\eta^2 = 0.008$). Fig. 1 presents the means of each condition.

Study 1 shows that when choice and decision autonomy are present, participants are more likely to adopt the AI-enabled technology in comparison to when these conditions are absent. In Study 2, we aim to replicate these findings while adding complexity as an independent variable.

3.2. Study 2: effects of choice and decision autonomy and complexity on adoption

In Study 2, we use an experimental approach to replicate the findings from Study 1, while controlling for the effect of product evaluation complexity on AI adoption. This is following the theories that consumers' have limited attentional and cognitive resources when processing complex information, which may increase the likelihood of using decision aids [13,50].

AI-enabled purchasing DSS tools are intended to help consumers make effective purchase decisions [13] and reduce consumers' product uncertainty [78]. Thus, if these aids can narrow down choices and make purchasing decisions on customers' behalf when product evaluation is complex, consumers may be more at ease with losing some of their autonomy. Complexity is reflected by the number of alternatives and attributes on which the alternatives are differentiated [79]. More attributes augment the amount of new information that needs to be evaluated to make final purchase decisions [80]. Consumers have limited attentional and cognitive resources; thus, they are selective in the information they process [50] and reduce decision complexity by relying on heuristics [81]. In addition, consumers exhibit over-dependence on algorithm-generated recommendations when complexity factors (limited attention, effort, and understanding) are present [50]. Therefore, in this study, we test if our findings from Study 1 (i.e., main effects of choice and decision autonomy on AI adoption) continue to hold for complex product evaluations.

3.2.1. Design, participants, and procedure

We use a 2 (choice autonomy: present vs. absent) x 2 (decision autonomy: present vs. absent) x 2 (complexity: high vs. low) between-subjects experimental design. As in Study 1, the manipulations involve a grocery shopping app, and the scenarios present the same manipulation of choice and decision autonomy. To manipulate complexity, the scenarios also describe the purchasing task as high or low in complexity. Consistent with prior research, we manipulated complexity by varying the number of product attributes [41,82]. Participants in the low complexity group read that they typically consider 3 important attributes when considering a purchase of a grocery item; those in the high complexity group instead read that they typically consider 20 important attributes. As in Study 1, we used participants' likelihood of adopting the AI-enabled app as the dependent variable. Appendix B presents details of the manipulations and measures used in the study, along with information about participant frequency per condition. A total of 563 Prolific panel members residing in the United States ($M_{Age} = 33.00$ years; 64.7% women) received monetary compensation for participating in the study.

3.2.2. Results and discussion

3.2.2.1. Manipulation checks. Participants assigned to the choice autonomy-present condition indicate greater control over their choices than those in the choice autonomy-absent condition ($M_{Choice} = 4.88$, SD $= 1.70$; $M_{NoChoice} = 3.31$, SD $= 1.57$; $t(561) = -11.42$, $p < .001$, *Cohen's d = 0.96*). Those assigned to the decision autonomy-present condition indicate greater control over their purchase decision autonomy ($M_{Decision} = 5.49$, SD $= 1.48$; $M_{NoDecision} = 3.38$, SD $= 1.79$; $t(561) = -15.24$, $p < .001$; *Cohen's d = 1.32*). For the complexity manipulation check, we asked participants to rate their level of agreement with two statements [79]: "Based on the scenario you read, how complex is it to choose a grocery item?" (1 = "not at all complex," 7 = "very complex") and "Based on the scenario you read, to what extent would you be overtaxed by the task of buying groceries?" (1 = "not at all overtaxed," 7 = "very overtaxed"). The two variables load into one factor ($\alpha = 0.74$; average loading $= 0.89$), so we checked the validity of our manipulation by using their average. The manipulation is successful: those assigned to the high complexity condition indicate a higher level of agreement than those in the low complexity condition ($M_{HighComplexity} = 4.23$, SD $= 1.52$; $M_{LowComplexity} = 3.29$, SD $= 1.45$; $t(561) = -7.52$, $p < .001$; *Cohen's d = 0.63*).

3.2.2.2. Adoption of AI. A three-way ANOVA reveals significant main effects of choice autonomy on AI adoption ($F(1, 555) = 5.92$, $p = .015$, $\eta^2 = 0.01$) and decision autonomy on AI adoption ($F(1, 555) = 29.34$, p

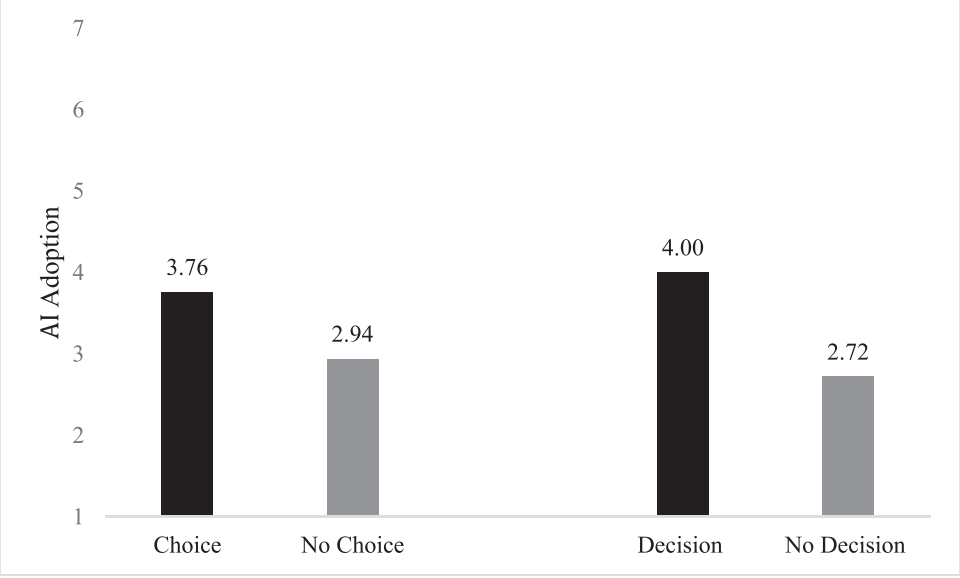


Fig. 1. Impact of choice and decision on AI adoption: Study 1.

$< .001$, $\eta^2 = 0.05$), which provides further support for H1. In contrast, the main effect of complexity is not significant ($F(1, 555) = 2.84$, $p = .09$, $\eta^2 = 0.005$). To ensure the robustness of our analysis, we also test for interaction effects between the independent variables. In contrast to Study 1, the interaction of choice and decision autonomy is not significant ($F(1, 555 = 1.78$, $p > .1$). The insignificant finding indicates that choice autonomy does not moderate the effect of decision autonomy on AI adoption and vice versa. Choice and decision autonomy affect AI adoption due to differing underlying reasons that are exclusive to each autonomy dimension. Perhaps, this is the reason that the absence or presence of one autonomy does not moderate the effect of another autonomy, at least not consistently. For example, choice autonomy allows customers to find ideal options that match their preferences [60] and fulfill their variety-seeking needs (e.g., [61]), while decision autonomy enables them to learn from their past decisions to enhance their long-term goals [64] and permits them to take ownership of their purchasing actions [68]. As choice autonomy and decision autonomy provide different benefits to customers' well-being, they do not necessarily interact to affect AI adoption. Due to the lack of underlying theoretical reasoning and the insignificant finding in Study 2, the significant interactive effect found in Study 1 should be interpreted with caution. Finally, the interaction of choice autonomy and complexity ($F(1, 555) = 1.24$, $p > .26$), and decision autonomy and complexity are not significant ($F(1, 555) = 1.93$, $p > .16$). The three-way interaction is equally insignificant ($F(1, 555) = 0.31$, $p > .86$). Fig. 2 presents the means of each condition.

Consistent with Study 1, these results provide evidence for the role of choice and decision autonomy as drivers of AI adoption. Participants indicated a higher likelihood of using the AI-enabled app if choice and decision autonomy are present (vs. absent). Furthermore, the main effect of complexity is not significant; the effects of choice and decision autonomy on AI adoption are also not moderated by complexity. Thus, Study 2 establishes that choice and decision autonomy underscore customers' likelihood of adopting AI.

3.3. Study 3: testing identity as a boundary condition

Study 2 provides further evidence that both choice and decision autonomy influence the adoption of AI-enabled technology, even after controlling for complexity of choice evaluation. To extend our understanding of the relationship between choice and decision autonomy with AI adoption, in Study 3, we test H2 to explore a circumstance in which these effects might be attenuated or augmented. Specifically, we examine identity salience as a boundary condition.

3.3.1. Design, participants, and procedure

In Study 3, we use a between-subjects experimental design: 2 choice autonomy (present vs. absent) x 2 decision autonomy (present vs. absent) x identity. The manipulations involve a running app that enables consumers to connect to an online running shoe retailer. In these scenarios, the role of the app is to ease the purchasing process of a product needed to perform the identity salient activity (i.e., running shoes). We measure the extent to which running constitutes customers' identity using a 3-item scale [39], and we calculate the average of the three items to create an identity score (44 $\alpha = 0.96$, $M = 3.09$; $SD = 1.70$). Appendix C presents details of the manipulations and measures, as well as the frequency of participants per condition. A total of 600 participants ($M_{Age} = 35.82$ years; 54.3% men) recruited from Prolific residing in the United States completed the study in exchange for monetary compensation.

3.3.2. Results and discussion

3.3.2.1. Manipulation checks. For manipulation checks, we use the measures from Study 1. As expected, participants assigned to the choice autonomy-present condition indicate they have greater control over their choices than those in the choice autonomy-absent condition ($M_{Choice} = 4.55$, SD = 2.19; $M_{NoChoice} = 2.68$, SD = 1.60; $t(598) = -11.96$, $p < .001$, *Cohen's d = 0.97*). Similarly, those assigned to the decision autonomy-present condition indicate greater control over their purchase decision than those in the decision autonomy-absent condition ($M_{Decision} = 5.83$, SD = 1. 42; $M_{NoDecision} = 2.46$, SD = 1.74; $t(598) = -25.95$, $p < .001$, *Cohen's d = 2.12*).

3.3.2.2. Adoption of AI. Using a bootstrapped sample (5000) procedure with PROCESS Model 2 [83], we estimate a regression model with "choice" and "decision" autonomy coded as 1 when present and 0 when absent, and the mean-centered identity as the continuous moderator. AI adoption is entered as Y (dependent variable), identity is entered as X, choice autonomy is entered as W, and decision autonomy is entered as Z. As Process Model 2 allows variable X to interact with variable W and variable Z simultaneously, it allows us to examine the interaction between identity and choice and the interaction between identity and decision. There is a main effect of identity on adoption (β =0.3209, SE =0.0655, CI95 = [0.1923, 0.4495]). We also find a main effect of decision on adoption (β =1.6393, SE =0.1336, CI95 = [1.3769, 1.9017]), while the main effect of choice on adoption is insignificant (β =0.0893, SE =0.1337, CI95 = [−0.1733, 0.3518]). In line with our conceptualization, we find a significant interaction effect between choice autonomy and identity on AI adoption (β = −0.1572, SE = 0.0788, CI95 = [−0.3120, −0.0025]; $F(1, 594) = 3.98$, $p = .04$) and a significant interaction effect between decision autonomy and identity on AI adoption (β = 0.1933, SE = 0.0789, CI95 = [0.0383, 0.3483]; $F(1, 594) = 6.00$, $p = .01$). Thus, H2 is supported.

Fig. 3 shows that when identity salience is low, the presence of choice autonomy leads to higher adoption relative to when choice autonomy is absent. When identity salience is high, the absence of choice autonomy leads to higher adoption relative to when choice autonomy is present. Fig. 4 shows that as identity salience increases, adoption increases in both decision absent and decision presence conditions. However, the moderating impact of identity salience on adoption is higher when decision autonomy is present than when decision autonomy is absent.

The results are in line with our theorizing that the AI-enabled purchasing app is relevant to customers' identity-related goals and allows internal attribution to occur. Identification with the activity increases customer adoption of the app among those who lack choice autonomy. The strong identification individuals prefer AI to narrow choices for them in comparison to those with lower identification. Identity salience positively moderates the decision autonomy–AI adoption relationship such that in both conditions (absence vs. presence), the stronger identity salience, the higher the likelihood of adoption. Thus, the absence of choice and decision autonomy as a deterrent to AI adoption is less prominent when identity salience is strong.[4]

[4] In a post-hoc study ($N = 443$; $M_{Age} = 26.43$ years; 55.5% women), we primed identity [84,85]. The study used a between-subjects experimental design: 2 choice autonomy (present vs. absent) x 2 decision autonomy (present vs. absent) x 2 identity (identity salient vs. control). We used AI adoption as the dependent variable and controlled for familiarity with this type of purchase situation. An ANCOVA analysis shows significant main effects of decision autonomy ($F(1, 434) = 8.87$, $p < .003$, $\eta2 = 0.02$) and identity salience on AI adoption ($F(1, 434) = 38.54$, $p < .001$, $\eta2 = 0.082$). Importantly, the interaction between choice and identity ($F(1, 434) = 5.19$, $p = .023$, $\eta2 = 0.012$) and the interaction between decision and identity ($F(1, 434) = 4.05$, $p = .045$, $\eta2 = 0.009$) are significant. The other effects are not significant (p's > 0.18). These findings replicate the ones from Study 3 and provide further support to our hypothesis.

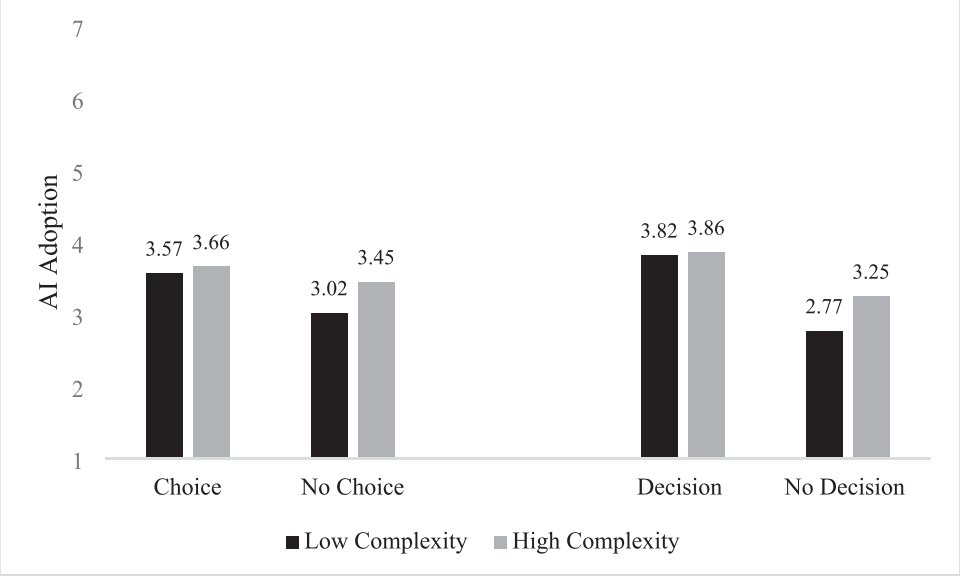


Fig. 2. Impact of choice, decision and complexity on AI adoption: Study 2.

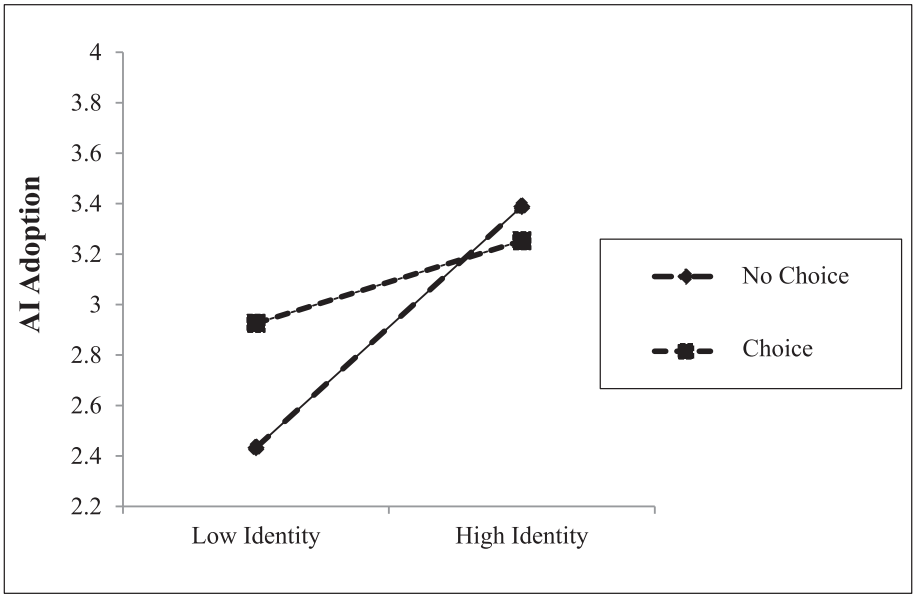


Fig. 3. Impact of choice and identity on AI adoption: Study 3.

3.4. Robustness checks

We re-ran the analyses for Study 1 and 2 using ANCOVA and Study 3 with PROCESS Model 2 by adding a control variable of prior experience using similar AI technology [86]. In Studies 1 and 2, we measured this control by asking participants if they currently use any kind of grocery shopping app (Yes = 1; No = 0). In Study 3, we asked participants to indicate if they have used other AI-enabled purchasing apps (e.g., Amazon Shopping, etc.) on their smartphone (Yes = 1; No = 0). The results of the estimation are consistent with existing analyses.

4. Discussion

In sum, Study 1 shows that when choice and decision autonomy are present, consumers are more likely to adopt the AI-enabled technology. Specifically, participants indicated higher likelihood of using the AI-enabled app if choice and decision autonomy are present (vs. absent). Study 2 replicates these findings even when complexity is controlled for. The main effect of complexity is not significant, and it does not moderate the effects of choice and decision autonomy on AI adoption. Study 3 tests the role of identity salience as a boundary condition of choice and decision autonomy on AI adoption. We found significant main effects of identity and decision autonomy on adoption, and significant interactions between choice autonomy and identity, and decision autonomy and identity on AI adoption. Finally, the absence of choice and decision autonomy as a deterrent to AI adoption is less prominent when identity salience is high.

4.1. Theoretical implications

Our research provides several theoretical contributions. To the best of our knowledge, this study is the first to outline a dimensional aspect of customer autonomy in AI-mediated purchase settings. We integrate extant knowledge of the consumer decision-making process [13,44,46]

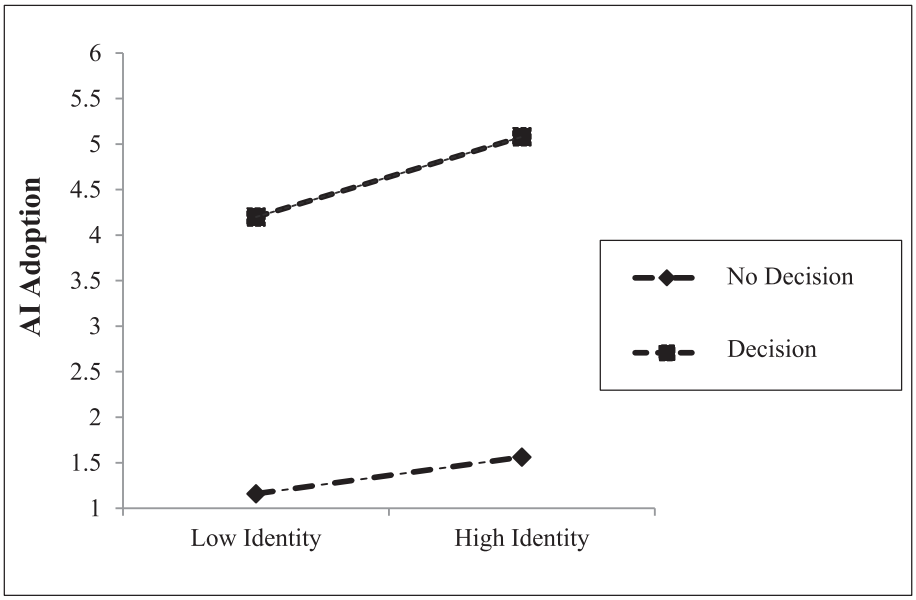


Fig. 4. Impact of decision and identity on AI adoption: Study 3.

with the application of AI in consumer purchase decisions. By identifying the type of task along the customer decision-making process that can be delegated to AI (i.e., choice set reduction and final choice decision), we determine two dimensions of consumer autonomy (i.e., choice and decision). We unpack the complexity of human autonomy in the purchase decision-making process in which customers can have full autonomy in both dimensions, no autonomy in either dimension, or full autonomy in just one dimension. In turn, we can measure each dimension's separate effect on AI adoption and grasp their relative value to customers. Hence, we propose that customer autonomy should be viewed across stages. By doing so, we offer an expanded conceptualization of consumer autonomy in AI-mediated purchase settings and shed light on the decision-making process in an AI era. Our findings are applicable even when customers are not aware that AI is applied behind the DSS. While customers may not realize that online retailers such as Amazon use AI to narrow down options in generating recommendations, they may notice that their consideration set is being restricted, leading to perceived lack of choice autonomy. Hence, customers may refuse to purchase from the retailer, signifying a rejection of the AI-enabled DSS.

Our work provides an empirical test to scholars' concern that a perception of diminishing autonomy (e.g., due to consumers' lack of control over the input underlying AI's algorithm [21,35]), has negative consequences [4,15,16]. We answered the call for more research examining whether perceived autonomy is valued more or less with the increasing prevalence of autonomous technologies [4]. AI-enabled DSSs are designed to help customers reduce decision difficulty, decision complexity, and preference uncertainty due to abundant choice options [80]. To date, although scholars predict that the benefits of AI may be outweighed by customers' autonomy concerns—which may affect AI evaluation and use—the empirical evidence has been lacking [15,47]. The findings of our studies in general and Study 2 in particular suggest that customers' desire to preserve their autonomy exceeds the need to reduce the time and cognitive effort resulting from complex decisions. Our work echoes scholars' warning that designing a system with a greater degree of autonomy must be done with clear understanding of its relative consequences to other actors, such as consumers [36]. Consumers may exercise their rights not to grant their autonomy to AI by refusing to use the technology. Thus, our research findings provide support for the call to design a system that fosters cooperation between humans and AI-based digital assistants, in which involving humans to an appropriate level in the task performance [87].

Finally, we identify a circumstance in which AI resistance is attenuated. Specifically, identity interacts with choice autonomy, such that when identity salience is high, the absence of choice autonomy leads to higher adoption relative to when choice autonomy is present. Furthermore, identity positively moderates the decision autonomy–AI adoption relationship; as identity salience increases, adoption increases in both decision absent and decision present conditions. These effects occur because AI-enabled purchasing tools complement consumers in achieving their identity-related goals while still allowing the attribution of identity-relevant activity outcomes to the self. They free consumers from the need to perform the task of repurchasing products essential to their identity-salient activity, allowing them to focus their time and effort on performing identity-salient activities. This finding adds to our understanding of the boundary conditions of customers' resistance to AI and enriches knowledge of the role of identity. Prior studies have focused on the delegation of tasks to automated products that perform identity-salient activities on customers' behalf; the more salient the identity, the higher their resistance to automated products because they can no longer attribute the task outcome to themselves [39]. In contrast, we focus on purchase delegation to AI involving items that are essential to identity-salient activity. Our results provide a new contribution by showing that the adoption of AI can increase if the technology complements efforts to realize an identity.

4.2. Managerial implications

4.2.1. Encouraging adoption through identity-salience enhancing function

The more consumers identify with a particular activity (e.g., running), the more likely they adopt an AI-enabled app that performs purchasing-related tasks pertinent to that activity (e.g., buying running shoes). This finding highlights a number of important implications for managers. First, to encourage adoption, the purchasing app should be offered for products that do not take over functions that are pertinent to their identity to allow internal attributions to occur. For example, an AI-purchasing app should be offered for running shoes and grocery shopping because the identity-salient activities themselves (e.g., running and cooking) are still performed by consumers. Second, it should be made clear to potential users that the purchasing app enhances their identities as they reduce their time and effort in purchasing products required to perform their identity. This way, their time and cognitive resources can be focused on their identity-salient activities. For example, running shoe

manufacturers could use a framing approach to promote the identity-enhancing function of the AI-enabled app: "We replace your running shoes so that you can spend your energy on what you love: running." Third, marketers should enhance identity salience of the activity in their promotional communication with firms. For example, in the case of the running app, using a visual or audio commercial, consumers can be primed to increase their identity salience as a runner by asking customers to recall how they feel before, during and after each run accompanied by audio and visual stimulus [84].

4.2.2. Improve perceptions of choice autonomy

Customers are largely unwilling to relinquish their choice autonomy. As highlighted in choice abundance literature [60–62], customers value the benefits of choice. Therefore, in addition to presenting AI-generated consideration sets, firms can give customers the option to view available choices. For example, Instagram provides options to customers on setting whose feeds they want to see rather than being solely determined by the algorithm. In addition, managers could make it easier for consumers to search for alternatives on their platforms. For example, streaming service firms may allow users to search movies based on their current mood, inspirations (e.g., names of icons, personality type, product release dates, periods [e.g., 1970s, Western, Victorian]). They also could allow customers to use multiple filters. With such tactics, firms can boost perceived autonomy, in addition to offering AI-predicted choices.

4.2.3. Offer optimal choices but leave final decisions to customers

Customers value the autonomy to make final decisions; to adopt an AI-enabled purchasing aid (e.g., an app), they need to feel satisfied with their levels of perceived autonomy. Accordingly, harnessing the power of AI to predict when customers need products and what their optimal choices are implies a caveat. First, the decision-making process should not be delegated entirely to AI. Instead, AI might offer predicted optimal choices but leave the final decisions to customers. Because our results are consistent across a variety of products, firms should carefully consider the impact of relinquishing customers' decision autonomy to AI such as in the case of ASS or shipping-then-shopping business models.

4.3. Limitations and future research directions

We acknowledge some limitations of this study that suggest opportunities for research. First, we focus on online purchases using scenario-based experiments; continued research could test our findings for AI-enabled purchasing aids using either a field study by launching an AI-enabled purchasing app or collecting secondary data by partnering with a firm that offers customers such technology. Furthermore, they may examine our findings in physical stores, such as changing rooms equipped with smart mirrors, in-store virtual assistants, or sales assistant robots [42]. Although in-store AI may perform the same functions discussed herein, the extent to which customers sense that their choice and decision autonomy are threatened may be attenuated by physical proximity and face-to-face interactions with the technology. Thus, it is possible that the physical presence of AI-empowered shopping assistants in-store alters customers' degree of perceived autonomy. Researchers might explore the generalizability of our findings to the use of AI in brick-and-mortar contexts and its potential role as a boundary condition.

Second, though we test the effects of choice and decision autonomy on AI adoption across multiple product categories, the effect might vary in others. In our studies, the proposed contexts were perceived by participants as more utilitarian rather than hedonic.[5] Since consumers have different perceptions of AI in different contexts (e.g. AI is perceived as more competent in providing recommendations in utilitarian than hedonic domains [41]), researchers could also test the impact of autonomy on AI adoption in hedonic situations. Perceived choice and decision autonomy might have stronger effects in product categories in which hedonic attributes are more salient. Future research may also want to test our findings in the utilitarian contexts covering products and services with greater decision-making involvement such as financial and health. Although in general consumer decision-making of utilitarian products follows the traditional cognitive and rational steps [56], it is possible that customers of financial and healthcare products are more inclined to expend their time and effort in searching for product choices and deliberating the alternatives due to the perceived risks and the potential lack of familiarity with them [88]. Thus, in these higher involvement utilitarian contexts, consumers may value their decision-making autonomy more than the utilitarian consumer products that we tested in our studies.

In conclusion, our research advances our understanding of various dimensions of autonomy at different steps of purchasing task processes and their perceived importance at the consumer–AI interface. We hope these findings stimulate further research into how to balance the primordial human need for autonomy against the practical need to perform tasks efficiently.

CRediT authorship contribution statement

Mariyani Ahmad Husairi: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. **Patricia Rossi:** Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

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Appendix A. Study 1: experimental design and manipulations

Below we describe the details of our experiments. We used no visual aids or stimuli other than the text described in the Appendices (A, B, and C). All the text presented to participants is described below, as they saw it on their screen. Except for the actual conditions (i.e., groceries vs. running shoes), Studies 1, 2 and 3 use the same setup, interface, and measurements.

[5] In each study, we measured the extent that participants perceive the product domain as either hedonic or utilitarian by asking a question using a 7-point bipolar scale. In Studies 1 and 2 (and 3), we asked: "When you think about grocery shopping (running shoes), how would you classify this activity (product)?" The questions were anchored on (1) Goal-oriented; (7) Pleasure-oriented. The mean across the three studies is 2.77.

[Introduction] Participants are asked to imagine that they have installed a running app on their phone that enables them to connect to an online shoe retailer. It reads as: "You installed an app of your most frequented grocery store on your phone. The app enables you to purchase your groceries online, which will be delivered to your address." Afterward, participants are randomly presented with one of the four scenarios below.

[0 choice autonomy, 0 decision autonomy, 135 participants*]* The retailer's machine learning algorithm can narrow down the options for groceries it assumes to be the most relevant for you. To make it easier for you, **you do not see all the grocery choices that are available**. Instead, when the app identifies that you have a need, it searches for an optimal choice, and **it executes the purchase on your behalf.**

[0 choice autonomy, 1 decision autonomy, 137 participants*]* The retailer's machine learning algorithm can narrow down the options for groceries it assumes to be the most relevant for you. To make it easier for you, **you do not see all the grocery choices that are available.** Instead, when the app identifies that you have a need, it searches for an optimal choice and makes a suggestion. **You then make the final choice and you execute the purchase yourself whenever you have a need.**

[1 choice autonomy, 0 decision autonomy, 132 participants*]* The retailer's machine learning algorithm **does not narrow down the grocery choices** it assumes to be the most relevant for you. This means that you can browse **ALL grocery choices that are available,** press the like button, and you can add them to your wish list. However, when the app identifies that you have a need, it searches for an optimal choice, and i**t executes the purchase on your behalf.**

[1 choice autonomy, 1 decision autonomy, 131 participants*]* The retailer's machine learning algorithm **does not narrow down the grocery choices** it assumes to be the most relevant for you. This means that you can browse **ALL grocery choices that are available**, press the like button, and you can add them to your wish list. However, when the app identifies that you have a need, it searches for an optimal choice and makes a suggestion for you. **You then make the final choice and you execute the purchase yourself whenever you have a need.**

Next, participants were asked to answer questions that measured the dependent variable, manipulation checks, and demographics (i.e., gender and age). Below are the measures:

Measures

- *Dependent variable:*
  - o Adoption of AI: How likely are you to use this app? (1 = Very unlikely; 7 = Very likely)
- *Manipulation checks* (based on [77]):

Based on the scenario you read:

- o How much control do you feel you have over the grocery choices you get to see? (1 = Do not have control at all; 7 = Have a lot of control)
- o How much control do you feel you have over the purchase decision of grocery products? (1 = Do not have control at all; 7 = Have a lot of control)

Appendix B. Study 2: experimental design and manipulations

Study 2 followed the same setup used in Study 1. Study 2 had a 2 (choice autonomy: present vs. absent) x 2 (decision autonomy: present vs. absent) x 2 (complexity: high vs. low) between-subjects experimental design. To manipulate choice autonomy (present vs. absent) and decision autonomy (present vs. absent), we used the same scenarios from Study 1 (i.e., the grocery shopping app). Below we present the manipulations used in the complexity (high vs. low) conditions. The manipulation used in each condition are in bold.

[Scenarios on choice autonomy and decision autonomy]

The scenarios on choice autonomy and decision autonomy were the same as Study 1. Participants were randomly presented with one scenario. After reading the scenario, participants were presented with one of the two complexity manipulations.

[Manipulation for low complexity] For each grocery item, you typically consider **3 important attributes** when making a purchase decision. They include **taste, packaging size, and price**.

[Manipulation for high complexity] For each grocery item, you typically consider **20 important attributes** when making a purchase decision. They include **brand name, taste, smell, texture, appearance, nutritional values, packaging size, the type of packaging (glass, bottle, paper, plastic), the attractiveness of the packaging, cooking/preparation time, price, place of origin, chemical composition, physical composition, expiration date, whether it contains preservatives, the type of fertilizer used, whether pesticides were used, the use of hormone and antibiotic additives and whether the manufacturer does not pollute the environment.**

Below we identify the number of participants per condition.

[0 choice autonomy, 0 decision autonomy, low complexity, 66 participants*]*

[0 choice autonomy, 1 decision autonomy, low complexity, 72 participants*]*

[0 choice autonomy, 0 decision autonomy, high complexity, 69 participants*]*

[0 choice autonomy, 1 decision autonomy, high complexity, 71 participants*]*

[1 choice autonomy, 0 decision autonomy, low complexity, 71 participants*]*

[1 choice autonomy, 1 decision autonomy, low complexity, 72 participants*]*

[1 choice autonomy, 0 decision autonomy, high complexity, 69 participants*]*

[1 choice autonomy, 1 decision autonomy, 0 explanation, 73 participants*]*

After reading the scenarios, participants are asked to complete the measures described below.

Measures

- *Dependent variable and manipulation checks for choice and decision autonomy:* the same as Study 1.
- *Manipulation checks for complexity* [79]:

Based on the scenario you read:

- o How complex is it to choose a grocery item? (1 = Not at all complex; 7 = Very complex)
- o To what extent would you be overtaxed by the task of buying groceries? (1 = Not at all overtaxed; 7 = Very overtaxed)

Appendix C. Study 3: experimental design and manipulations

Study 3 followed the same setup used in our previous studies. No visual aids were provided to participants. Then, they were presented with the following introduction.

[Introduction] You will be presented with a consumption scenario. Please immerse yourself in the scenario as much as possible.

You installed a running app on your phone. The app enables you to connect to an online running shoes retailer.

Then participants are randomly presented with one of the four scenarios below.

[0 choice, 0 decision, 151 participants*]* Because the running app records your running habits, the online retailer's machine learning algorithm can predict when you have a need for new running shoes. To make it easier for you, **you do not see all the running shoes choices that are available**. Instead, once it identifies that you have the need, **it searches**

for an optimal choice, and **it executes the purchase on your behalf.**

[0 choice, 1 decision, 149 participants*]* Because the running app records your running habits, the online retailer's machine learning algorithm can predict when you have a need and narrow down running shoes it assumes to be the most relevant for you. To make it easier for you, **you do not see all of the running shoes choices that are available**. Instead, **it searches for an optimal choice** once it identifies you have a need. **You then make the final choice and you execute the purchase yourself** whenever you have the need.

[1 choice, 0 decision, 150 participants*]* The online retailer's machine learning algorithm **does not narrow down** running shoes it assumes to be the most relevant for you. This means **you can see all of the running shoes choices that are available** on the retailer's website. However, because the running app records your running habits, the online retailer's machine learning-algorithm can predict when you have a need for new running shoes. To make it easier for you, once it identifies that you have a need, it searches for an optimal choice based on your browsing, clicking, wish list and purchase history, and **it executes the purchase on your behalf**.

[1 choice, 1 decision, 150 participants*]* The running app records your running habits. However, the online retailer's machine learning algorithm **does not narrow down** running shoes it assumes to be the most relevant for you. This means, **you can see all of the running shoes choices that are available** on the retailer's website. **You then make the final choice and you execute the purchase yourself** whenever you have the need for a new pair of running shoes.

After reading the scenarios, participants were asked to complete the measures described below.

Measures

- *Identity* [39]: Please rate your agreement with the following statements (1 = Strongly disagree; 7 = Strongly agree, α = 0.96)
  - o I am really into running.
  - o I identify myself as a runner.
  - o Running is one of my favorite hobbies.
- *Dependent variable and manipulation checks*: the same as Study 1

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